



Lecture 33-EtherChannel (Port Channel)



What is EtherChannel?



EtherChannel (also called Port Channel / LAG - Link Aggregation Group) allows you to combine multiple physical interfaces into **one logical interface**.



Simple Definition:

- Multiple cables → act like **ONE cable**
- Multiple ports → behave like **ONE port**



Types of EtherChannel

◆ 1. Layer 2 EtherChannel

- Group of **switch ports**
- Works like a **single Layer 2 interface**
- Used for switching (VLANs, trunks)

◆ 2. Layer 3 EtherChannel

- Group of **routed ports**
- Works like a **single Layer 3 interface**
- You can assign an **IP address** directly



Real-Life Example

Imagine a **highway** :

- One road = slow traffic 
- Multiple lanes = faster traffic 

- 👉 EtherChannel = **multiple lanes combined as one big road**
 - 👉 More speed + no traffic jam = better performance 🚀
-

! Problem Without EtherChannel

◆ Oversubscription

- End devices bandwidth > switch uplink bandwidth
- Leads to **congestion** 😞

◆ STP Issue

- If multiple links exist:
 - STP blocks all except ONE ❌
 - Only one link is used 😞

🔥 Why?

- To prevent **Layer 2 loops**
 - Otherwise → **Broadcast storm** 💣
-

✅ Solution: EtherChannel

- ✓ All links are ACTIVE (no blocking)
 - ✓ STP treats them as ONE interface
 - ✓ Traffic is **load balanced**
 - ✓ No loops
-

```
ASW1(config-if-range)#channel-group 1 mode ?
active      Enable LACP unconditionally
auto        Enable PAgP only if a PAgP device is detected
desirable   Enable PAgP unconditionally
on          Enable Etherchannel only
passive     Enable LACP only if a LACP device is detected

ASW1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
```

EtherChannel Protocols

◆ 1. PAgP (Port Aggregation Protocol)

- Cisco proprietary 
- Max: **8 links**

Modes:

Mode	Meaning
------	---------

auto	passive
------	---------

desirable	active
-----------	--------

✓ Working:

- desirable + auto = 
- desirable + desirable = 
- auto + auto = 

◆ 2. LACP (Link Aggregation Control Protocol)

- Industry standard  (IEEE 802.3ad)
- Max: **16 links (8 active + 8 standby)**

Modes:

Mode	Meaning
------	---------

active	active
--------	--------

passive	passive
---------	---------

✓ Working:

- active + passive = 
- active + active = 
- passive + passive = 

◆ 3. Static EtherChannel

- No protocol ✗
- Manual configuration

✓ Only works:

- on + on = ✓
- on + anything else = ✗

Load Balancing in EtherChannel

◆ Based on "Flows"

- A **flow** = communication between two devices

👉 Same flow → same cable

👉 Different flows → different cables

⚠️ Why?

To avoid **packet out-of-order problem**

Load Balancing Methods

- Source MAC
 - Destination MAC
 - Source + Destination MAC
 - Source IP
 - Destination IP
 - Source + Destination IP
-

Check Load Balancing

show etherchannel load-balance

Change Method

conf t

port-channel load-balance src-dst-mac

Configuration of EtherChannel

◆ PAgP Configuration

interface range g0/0 - 3
channel-group 1 mode desirable

✓ Creates:

Port-channel 1

◆ LACP Configuration

interface range g0/0 - 3
channel-group 1 mode active

◆ Static Configuration

interface range g0/0 - 3
channel-group 1 mode on

Important Rule

✓ Channel-group number:

- MUST match on same switch
 - DOES NOT need to match on other switch
-

Manual Protocol Selection

```
channel-protocol lacp  
channel-group 1 mode active
```

 If mismatch → ERROR

Configure Port-Channel Interface

```
interface port-channel 1  
switchport trunk encapsulation dot1q  
switchport mode trunk
```

Running Config Example

- Physical interfaces auto-configured from port-channel

```
interface GigabitEthernet0/0  
channel-group 1 mode active
```

Important Conditions

All member interfaces **MUST** have:

- ✓ Same speed
- ✓ Same duplex
- ✓ Same VLAN
- ✓ Same trunk/access mode

 Otherwise → interface excluded

Verification Commands (VERY IMPORTANT)

✓ 1. Show Summary

show etherchannel summary

Output Meaning:

Symbol Meaning

S Layer 2

R Layer 3

U In use

P Bundled

s Suspended ✗

D Down ✗

✓ Correct:

- L2 → SU
- L3 → RU

✓ 2. Port-Channel Info

show etherchannel port-channel

✓ 3. Trunk Check

show interfaces trunk

✓ 4. IP Check

show ip interface brief

✓ 5. Running Config

```
show running-config
```

✓ 6. Load Balance

```
show etherchannel load-balance
```



Layer 3 EtherChannel Configuration

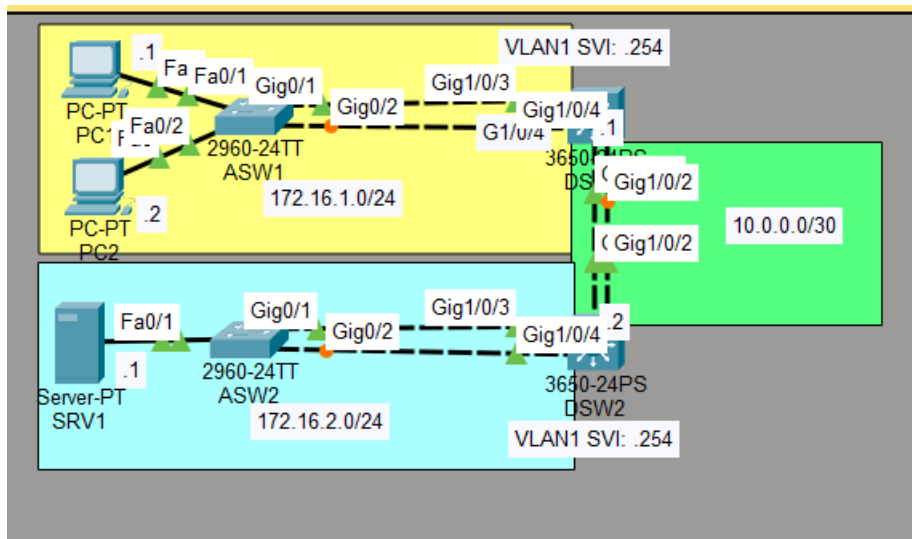
```
interface range g0/0 - 3  
no switchport  
channel-group 1 mode active
```

Then:

```
interface port-channel 1  
ip address 10.0.0.1 255.255.255.252
```

✓ IP is assigned on **Port-channel**, not physical ports

◆ Topology



- ASW1 ↔ DSW1 → LACP
- ASW2 ↔ DSW2 → PAgP
- DSW1 ↔ DSW2 → Layer 3 Static
- PCs → ASW1
- Server → ASW2

✓ **1. Configure Layer 2 EtherChannel between ASW1 and DSW1 using LACP. Configure it as a trunk.**

◆ On ASW1

```
conf t
interface range g0/1 - 2
switchport mode trunk
channel-group 1 mode active
exit
```

```
interface port-channel 1
switchport mode trunk
do wr
```

◆ **On DSW1**

```
conf t
interface range g1/0/3 - 4
switchport mode trunk
channel-group 1 mode passive
exit
```

```
interface port-channel 1
switchport mode trunk
do wr
```

✓ **active + passive = LACP works**

✓ **2. Configure Layer 2 EtherChannel between ASW2 and DSW2 using PAgP.**

Configure it as a trunk.

◆ **On ASW2**

```
conf t
interface range g0/1 - 2
switchport mode trunk
channel-group 2 mode desirable
exit
```

```
interface port-channel 2
switchport mode trunk
do wr
```

◆ On DSW2

```
conf t
interface range g1/0/3 - 4
switchport mode trunk
channel-group 2 mode auto
exit
```

```
interface port-channel 2
switchport mode trunk
do wr
```

✓ desirable + auto = PAgP works

✓ 3. Configure Layer 3 EtherChannel between DSW1 and DSW2 using static EtherChannel.

◆ On DSW1

```
conf t
interface range g1/0/1 - 2
no switchport
channel-group 3 mode on
exit
```

```
interface port-channel 3
no switchport
ip address 10.0.0.1 255.255.255.252
end
```

◆ On DSW2

```
conf t
interface range g1/0/1 - 2
no switchport
channel-group 3 mode on
exit
```

```
interface port-channel 3
no switchport
ip address 10.0.0.2 255.255.255.252
end
```

✓ Static EtherChannel → on + on only works

✓ 4. Configure routes to allow the PCs to reach SRV1.

- Server network behind DSW2
- PCs behind DSW1

◆ On DSW1

```
ip route 172.168.1.0 255.255.255.0 10.0.0.2
```

◆ On DSW2

```
ip route 172.16.2.0 255.255.255.0 10.0.0.1
```

✓ This allows both sides to communicate

✓ 5. What is the default EtherChannel load-balancing method used on each switch?

Check on any switch:

```
show etherchannel load-balance
```

✓ Default output usually:

```
src-dst-ip
```

👉 Meaning:

Source XOR Destination IP

✓ **6. Configure the switches to load-balance based on source and destination IP addresses.**

◆ **On ALL switches**

```
conf t
port-channel load-balance src-dst-ip
end
```

Verify:

```
show etherchannel load-balance
```

🔍 **Verification Commands (VERY IMPORTANT)**

```
show etherchannel summary
show etherchannel port-channel
show interfaces trunk
show ip interface brief
show spanning-tree
```

Output flags:

- **SU** → Layer 2 + in use ✓
- **RU** → Layer 3 + in use ✓
- **P** → bundled port ✓
- **s** → suspended ✗

⚠ **Common Mistakes**

- ✗ VLAN mismatch
 - ✗ Trunk/Access mismatch
 - ✗ Speed/Duplex mismatch
 - ✗ Wrong mode combination
 - ✗ Using "on" with LACP/PAgP
-

Quick Revision Table

Type	Protocol	Modes	Works
LACP	active/passive	active + passive	✓
PAgP	desirable/auto	desirable + auto	✓
Static	on	on + on	✓

Final Summary

- ✓ EtherChannel = Multiple links → One logical link
 - ✓ Prevents STP blocking
 - ✓ Improves speed + redundancy 
 - ✓ Load balancing using flows
 - ✓ LACP (standard) > PAgP (Cisco only)
-

EtherChannel Summary 🚀

🧠 Core Concept

👉 EtherChannel = **Multiple physical links combined into ONE logical link**

- ✓ Multiple cables → ONE logical interface
 - ✓ Multiple ports → ONE port
 - ✓ STP treats it as **single link**
-

🎯 Why EtherChannel is Needed

✗ Without EtherChannel:

- STP blocks redundant links 😞
- Bandwidth waste
- Congestion (Oversubscription)

✓ With EtherChannel:

- All links ACTIVE 🔥
 - Load balancing ⚖️
 - No STP blocking
 - High speed + redundancy 🚀
-

🧩 Types of EtherChannel

Type	Description
------	-------------

Layer 2	Works like switch port (VLAN, trunk)
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Layer 3	Routed port with IP address
---------	-----------------------------

Protocols

Protocol Type		Key Point
LACP	Standard (IEEE 802.3ad)	Most used 🌐
PAgP	Cisco only	Old
Static	Manual	Risky

Load Balancing

- ✓ Based on **flows (not packets)**
- ✓ Prevents packet disorder

Methods:

- MAC based
 - IP based (most common)
-

Important Conditions

All interfaces must have:

- ✓ Same speed
 - ✓ Same duplex
 - ✓ Same VLAN
 - ✓ Same trunk/access mode
-

Verification Commands

```
show etherchannel summary
show interfaces trunk
show ip interface brief
show etherchannel load-balance
```

Conclusion

 EtherChannel is used to:

- Increase bandwidth 
- Provide redundancy 
- Avoid STP blocking 
- Improve performance 

 **Best Practice:**

 Use **LACP (active/passive)** in real networks

 Final one-liner:

“EtherChannel combines multiple physical links into one logical link to increase bandwidth and provide redundancy.”

Q & A

? Q1: What is EtherChannel?

✓ Answer:

EtherChannel is a technology that combines multiple physical interfaces into a single logical interface, allowing increased bandwidth and redundancy while preventing STP from blocking links.

? Q2: Why do we use EtherChannel?

✓ Answer:

We use EtherChannel to:

- Increase bandwidth 🚀
 - Prevent STP blocking ❌
 - Provide redundancy 🔁
 - Improve network performance ⚡
-

? Q3: What is LAG?

✓ Answer:

LAG (Link Aggregation Group) is another name for EtherChannel.

? Q4: What problem does EtherChannel solve?

✓ Answer:

It solves:

- STP blocking issue
 - Oversubscription problem
 - Bandwidth limitation
-

⚙️ PROTOCOL QUESTIONS

❓ Q5: Difference between LACP and PAgP?

✅ Answer:

Feature LACP		PAgP
Type	Standard	Cisco only
IEEE	802.3ad	No
Usage	Modern networks	Legacy

❓ Q6: LACP modes?

✅ Answer:

- active (initiates)
- passive (waits)

✓ active + passive = works

✗ passive + passive = fails

❓ Q7: PAgP modes?

✅ Answer:

- desirable (active)
- auto (passive)

✓ desirable + auto = works

✗ auto + auto = fails

? Q8: Static EtherChannel?

✓ Answer:

- Manual configuration
 - No negotiation
 - ✓ on + on works
 - ✗ mismatch fails
-

? Q9: Configure LACP EtherChannel

✓ Answer:

```
interface range g0/0 - 3
channel-group 1 mode active
```

? Q10: Configure trunk on port-channel

```
interface port-channel 1
switchport mode trunk
```

? Q11: Layer 3 EtherChannel config

```
interface range g0/0 - 3
no switchport
channel-group 1 mode active
```

```
interface port-channel 1
ip address 10.0.0.1 255.255.255.252
```

? Q12: How to verify EtherChannel?

show etherchannel summary

✓ Look for:

- SU → Layer 2 working
 - RU → Layer 3 working
 - P → bundled
-

? Q13: How does load balancing work?

✓ Answer:

- Based on flows (not packets)
- Same flow → same link
- Different flows → different links

👉 Prevents packet reordering

? Q14: Why STP does not block EtherChannel?

✓ Answer:

Because STP sees EtherChannel as a single logical interface.

? Q15: Max links in LACP?

✓ Answer:

- 16 total
 - 8 active + 8 standby
-

? Q16: What happens if configuration mismatch?

✓ Answer:

- Ports go to **suspended state**
 - EtherChannel fails ✗
-

! TROUBLESHOOTING QUESTIONS

? Q17: EtherChannel not forming – reasons?

✓ Answer:

Common issues:

- VLAN mismatch ✗
 - Trunk mismatch ✗
 - Speed mismatch ✗
 - Duplex mismatch ✗
 - Wrong mode ✗
-

? Q18: What is suspended port?

✓ Answer:

Port is not added to EtherChannel due to mismatch configuration.

? Q19: Command to check load balancing?

show etherchannel load-balance

? Q20: Change load balancing method

port-channel load-balance src-dst-ip

SCENARIO-BASED QUESTIONS

? Q21: Why only one link used without EtherChannel?

✓ Answer:

Because STP blocks redundant links to prevent loops.

? Q22: What happens if one link fails in EtherChannel?

✓ Answer:

- Traffic shifts to remaining links
 - No downtime 🔥
-

? Q23: Can we mix FastEthernet and Gigabit?

✗ Answer:

No, all interfaces must have same speed.

? Q24: Can we use different VLANs on member ports?

✗ No — must be identical

? Q25: Difference between Layer 2 and Layer 3 EtherChannel?

✓ Answer:

Feature	Layer 2	Layer 3
Works on	MAC	IP
VLAN support	Yes	No
IP address	No	Yes

? Q26: What is flow-based hashing?

✓ Answer:

Algorithm used to decide which link carries traffic.

? Q27: What is port-channel interface?

✓ Answer:

Logical interface representing all bundled physical ports.

? Q28: How to create EtherChannel between switches?

Steps:

- 1 Select interfaces
 - 2 Configure mode
 - 3 Create port-channel
 - 4 Apply trunk
-

? Q29: How to verify trunk in EtherChannel?

show interfaces trunk

? Q30: What happens if one side is LACP and other is PAgP?

✗ EtherChannel will NOT form

🚀 One Liner

“Explain EtherChannel in simple terms”

100 Answer:

👉 “EtherChannel combines multiple physical links into one logical link to increase bandwidth, provide redundancy, and prevent STP from blocking links.”
